

Supervised Learning - Study Notes

What is supervised learning?

Supervised learning is a type of machine learning in which a model learns from labeled examples. Each training example consists of input features and a known target, also called a label. The goal is to learn a mapping from inputs to targets that generalizes to new, unseen data.

Classification vs regression

Classification predicts a discrete category, for example whether an email is spam or not spam. Regression predicts a continuous numeric value, for example the price of a house. Both use labeled data; the only difference is the type of target the model has to predict.

Common algorithms

Linear regression fits a straight line (or a hyperplane in higher dimensions) to predict a numeric target. Logistic regression is used for binary classification and outputs a probability between 0 and 1. Decision trees split the data with a sequence of if-else questions about the features. Random forests train many decision trees on random subsets of the data and features and then average or vote on their predictions, which reduces overfitting. k-Nearest Neighbors (k-NN) predicts using the labels of the k closest training examples. Support Vector Machines (SVM) find the decision boundary that maximizes the margin between classes.

Loss functions

Training a supervised model means minimizing a loss function that measures prediction error. Mean squared error (MSE) is the standard loss for regression problems. Cross-entropy, also called log loss, is the standard loss for classification problems.

Typical workflow

First collect and label the data. Then split it into training, validation and test sets. Choose a model, train it on the training set, tune hyperparameters on the validation set, and report the final performance on the test set only once, at the very end.

Practical examples

Spam detection and image recognition are classification tasks. House price prediction and demand forecasting are regression tasks. Other common applications include credit scoring, medical diagnosis support and customer churn prediction.

Strengths and limitations

Supervised learning is powerful when many high-quality labels are available. Its main limitation is that labeling data is often slow and expensive, and noisy labels lead directly to a noisy model. A supervised model can only be as good as the data it was trained on.